

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - SIH26124
- Problem Statement Title- AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet
- Theme- Smart Automation
- PS Category- Software
- Team ID- SNPSU0161
- Team Name - SNPSU Flash Forge

Travillox — turning public transport into a mobile urban sensing network



Travillox Urban Sense

From navigation app → city-scale urban intelligence platform

Problem Description :

- Public buses already travel fixed routes across the city every day, but this daily movement is not used for any road or infrastructure monitoring.
- Potholes, damaged roads, waterlogging and missing infrastructure are mostly caught only through manual inspection or citizen complaints.
- Authorities have no continuous, geotagged, severity-ranked view of road and service issues to plan maintenance or respond quickly.
- Impact → Delayed repairs, reduced commuter safety, and reactive rather than proactive urban management.

Solution :

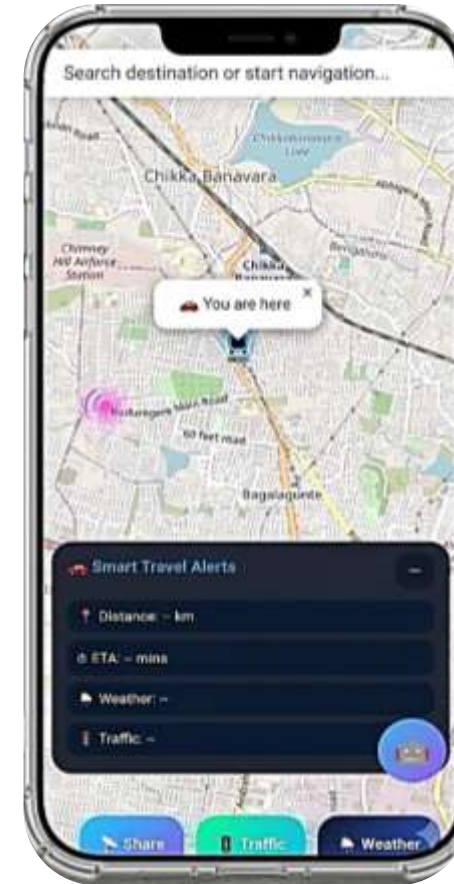
- Mobile sensing: Use cameras on public buses as moving observation points across regularly travelled city roads.
- AI understanding: Computer vision detects potholes, damaged roads, waterlogging, missing infrastructure and traffic conditions.
- Actionable intelligence: Geotag events, combine repeated sightings and feed verified road intelligence to Travillox and authorities.
- Impact → Travillox Urban Sense turns a navigation app into a city-scale urban intelligence platform.

Tech Stack :

- **Mapping & Navigation:** Leaflet, MapLibre GL 3D — animated route-flow engine
- **Computer Vision:** OpenCV, Ultralytics / YOLO (real-time object detection), RDD2022 road-damage detection methods
- **Geospatial Data:** OpenStreetMap, PostGIS (spatial database & geospatial queries)
- **Platform:** Installable PWA with offline app shell

Process :

- Bus camera + GPS — capture live video & location from moving buses
- Edge / Video AI — computer vision detects potholes, damage, waterlogging & traffic
- Geo-tag + Validation — geotag events and confirm via repeated sightings
- Urban Intelligence DB — store verified, structured road & issue intelligence
- Travillox Navigation — feed intelligence into the rider-facing app
- Authority Dashboard — surface issues, hotspots, SLA & resolution tracking
- Actionable Insights — prioritised alerts and evidence-based city decisions



Core architecture



Feasibility :

- Already built and working, not a concept — navigation, live-safety and smart-travel features are functioning today.
- Built on Leaflet + MapLibre GL 3D, with an animated route-flow engine already running.
- Uses open, well-documented tools — OpenStreetMap, OpenCV, Ultralytics/YOLO, PostGIS — keeping cost and licensing overhead low.
- Sensing layer reuses existing public-bus movement, so no separate fleet or hardware network needs to be built.

Solutions :

- Prioritise high-traffic and hotspot routes first, then expand coverage as more buses are fitted.
- Combine repeated sightings with severity tracking (Critical/High/Medium/Low) to validate detections and cut false positives.
- Resolution funnel (Reported → Acknowledged → In progress → Resolved) with a 72h SLA for High/Critical issues keeps authorities accountable.
- Installable PWA with offline app shell keeps the rider app usable even with patchy connectivity.

Challenges :

- Coverage is limited to the routes and stops the monitored buses already travel (42 routes / 318 stops today).
- Camera-based detection can be affected by low light, weather, or obstructed views.
- Value depends on authorities acting on the intelligence surfaced through the dashboard.

Apps Used



The Best Apps For
Travel

TRAVILLOX APP FEATURES

Already built and working, not a concept

Navigation & AI

- Turn-by-turn voice guidance
- AI chatbot for weather, traffic & safety
- 3D drive-mode chase camera
- Dark mode + destination autocomplete

Live Safety

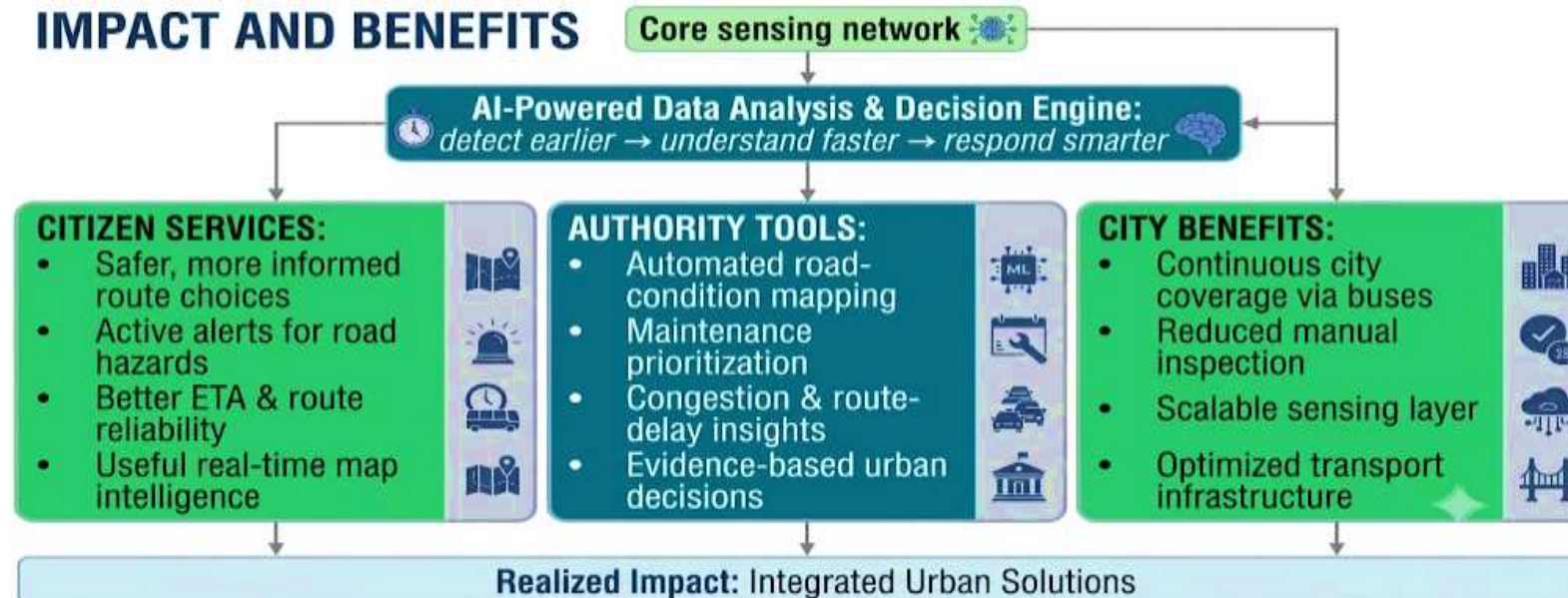
- PulseX one-tap emergency trigger
- Live location sharing via viewer link
- Live speed & position broadcast
- In-app feedback / comment channel

Smart Travel

- Weather-aware ETA calculations
- Live traffic layer + auto-reroute
- Smart ETA engine (speed + traffic aware)
- Installable PWA with offline app shell

Built on Leaflet + MapLibre GL 3D, with an animated route-flow engine already running

IMPACT AND BENEFITS



Primary reference :

- Smart India Hackathon 2026 — Problem Statement SIH26124
- Provided SIH 2026 Idea Presentation Template

Technical references :

- OpenStreetMap — open geospatial/map data
- OpenCV — computer-vision processing
- Ultralytics / YOLO documentation — real-time object detection
- RDD2022 / road-damage detection research — road defect datasets and methods
- PostGIS — spatial database and geospatial queries

Links :

- <https://www.sih.gov.in/sih2026PS>
- <https://www.openstreetmap.org/>
- <https://opencv.org/>
- <https://docs.ultralytics.com/>